DXCONNECTION

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CERTIFICATE OF CALIBRATION

Prepared for:
Lee Bussy
1237 Eagle
Hillsboro KS 67063-8071
United States

Certificate No: DXC-2026-3339-001

INSTRUMENT INFORMATION

Inst. Description: Power Meter
Manufacturer: Hewlett-Packard
Model: 437B
Serial Number: 3125U33223
Cust. ID. No.: N/A
Accuracy: MM

Calibration Site: In Shop
Date Received: N/A
Date of Cal.: 15-Oct-2026
Next Cal. Due: 15-Oct-2028
Proc.: MM
Calibration Result: Passed

As Received	As Left
	In Tolerance
Operational	Operational
	Full Calibration

STANDARDS USED

Model No.	Manufacturer	Description	Cal Date Due Date	Uncertainty
5100B	Fluke	Calibrator	21-Nov-2026	MM
3458A	Keysight	8-1/2 Dig Multimeter	20-Dec-2026	+/- 0.2%
11683A	Hewlett-Packard	Range Calibrator	12-Oct-2027	+/- 0.25%
432A	Hewlett-Packard	Power Meter	12-Oct-2027	+/- 2.0%
478A-H75	Hewlett-Packard	Thermistor Mount	12-Oct-2027	+/- 2.0%
5350B	Hewlett-Packard	Frequency Counter	22-Jul-2027	$< 1 \times 10^{-9}$

TEST ROOM ENVIRONMENT

Temperature: 21.3C / 70.4 F Relative Humidity: 23.0% Barometric Pressure: N/A in. Hg

V P I C S certifies that the above listed instrument meets or exceeds all published specifications. It has been calibrated using standards whose accuracies are traceable to the National Institute of Standards and Technology. Alternatively, accuracies have been derived from accepted values of natural physical constants or have been derived by the ratio type of self-calibration techniques. This certificate may be reproduced in full. Partial reproduction may be done only with written prior consent of V P I C S. Calibration performed using procedures and standards traceable to the National Institute of Standards.

Measurements certified by: David R. Smith ***Date:*** 10/15/2026

Title: Sr Engr

Hewlett-Packard
437B Power Meter

Adjustments and Calibration

Serial Number: 3125U33223 Date 15 Oct 2026

All measurements were taken with an Agilent 44301A Digital Voltmeter (DVM), which is currently in calibration.

1. Power Supply checks
   a. +5.0 VDC buss (+/- .25 V) 4.498 VDC

   b. +5.0 VDC buss - AC noise and ripple voltage (5mV RMS max) 0.77 mV (rms)

   c. -15.0 VDC buss (+/- .5 VDC) -15.158 VDC

   d. -15.0 VDC buss - AC noise and ripple voltage (5mV RMS max) 1.66 mV (rms)

   e. +15.0 VDC buss (+/- .5 VDC)
   15.101 V

   f. +15.0 VDC buss - AC noise and ripple voltage (5mV RMS max) 1.58 mV (rms)

   g. Calculate the difference between the absolute values of the -15.0 V and +15.0 V power supplies. The difference must be no greater than +/- 2.00% (-300 mv to +300 mv)

15.158 (Absolute value of +15.0 V)

15.101 (Absolute value of -15.0V)

57 mV (-300 mV < calculated value < +300 mV)

2. 220 Hz Frequency Adjustment – A4R87
   X
3. Range 4 Shaper Adjustment - A4R111 X
4. Range 5 Shaper Adjustment X
5. RAM Battery Check 3.448 VDC

Replaced BT1 RAM battery 8 Aug 2026

6. Power Reference Oscillator Frequency Adjustment – A4L5 __X__

7. Power Reference Oscillator Level Adjustment – G1A1R4 __X__

8. Functional Self Tests:

   a. ROM Test Pass
   b. RAM Test Pass
   c. Display Test Pass
   d. A/D Converter Test Pass
   e. Key's Test Pass
   f. All Test
      Pass

9. HPIB
   a. HPIB Address 13
   b. HPIB Functional Test Pass

10. Analog Meter Check ___X___

11. Calibration Completed ___X___

12. Settings:
    a. Special Status
       i. Sensor: 0
          1. ID Default
          2. Clr Tbl: Y
          3. Ref CF: 100.0%
          4. 00.00 Mz 100.0%
       ii. Limits: OFF
       iii. Auto Filter: 128
    b. Software: 2.0

David R. Smith — Test performed by

10-15-2026 — Date

Hewlett-Packard
437B Power Meter

Performance Test Record

Serial Number: 3125U33223 Date: 15-Oct-2026

Tested By D. Smith

Paragraph No.	Test	Min. Result	Actual Result	Max. Result
4-7	**ZERO CARRYOVER**			
	Power Meter Range			
	1	-0.05 uW	-0.01	0.05 uW
	S	-0.1 uW	0.0	0.1 uW
	3	0.001 mW	0.000	0.001 mW
	4	0.01 mW	0.00	0.01 mW
	5	0.1 mW	0.0	0.1 mW
4-8	**INSTRUMENT ACCURACY**			
	Accuracy	3.10 uW	3.16	3.23 uW
		9.90 uW	10.05	10.10 uW
		31.4 uW	31.6	31.8 uW
		99.5 uW	100.0	100.5 uW
		0.314 mW	0.316	0.318 mW
		0.995 mW	1.000	1.005 mW
		3.14 mW	3.16	3.18 mW
		9.95 mW	9.99	10.05 mW
		31.4 mW	31.6	31.8 mW
		99.5 mW	100.1	100.5 mW
4-9	**POWER REFERENCE**			
	Prf	0.988 mW	1.001	1.012 mW
	Freq. MHz			

Certified BY: David R. Smith